Ram Calculus for Rokhlin's Multiple-Mixing Problem

M280: Entropy, Conditional Screening, and Physical Residual Certificates

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Abstract

The September 23, 2026 manuscript on Rokhlin's problem claims that ordinary mixing of one invertible probability-preserving transformation implies mixing of every finite order. We examine its exact quantifiers, its reduction from arbitrary spaces to a standard zero-entropy witness, and the roles of ordered arrays, saturated extensions, operator squares, measurable labels, and finite blocks. The local Ram source TR-item-313 addresses this same core problem through actual off-diagonal laws and quantitative homoclinic residuals; it does not supply the manuscript's structural theorem.

Concrete refinements proved here include the sharp information inequality $J(A) \le I(W;C)/n$ for dissociated projective characters, and hence $A \le \sqrt{2H(W)/n}$ for finite words; an L^1 extension of local orbit cancellation; a finite van der Corput budget; square-root screening estimates; a Bessel fibre majorant; and signed homoclinic certificates with physical Gram-tail control. These are local results or explicitly conditional consequences, not independent certification of a universal solution. The entire pinned mathematical source was read. No specific fatal error was identified in this bounded review, but the infinite structural construction has not been independently reconstructed in full.

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1 The Exact Problem and the Status of This Report

Let T be an invertible, probability-preserving transformation of $(X, \mathcal{B}, \mu)$. We count the number of observed factors. Ordinary mixing means

$$\mu(A \cap T^{-n}B) \longrightarrow \mu(A)\mu(B) \quad \text{as } |n| \longrightarrow \infty$$

for each fixed pair of measurable sets. Mixing of order k means

$$\int_X \prod_{i=0}^{k-1} f_i(T^{t_i}x)\,\mathrm{d}\mu(x) \longrightarrow \prod_{i=0}^{k-1} \mu(f_i), \qquad \min_{1\le i<k}(t_i - t_{i-1}) \longrightarrow \infty, \tag{1.1}$$

for every fixed k, every fixed bounded $f_0, \dots, f_{k-1}$, and every increasing integer layout $0 = t_0 < t_1 < \cdots < t_{k-1}$. Indicators and bounded functions give equivalent formulations by simple-function approximation. Translation, sorting, and finitely many permutations recover the pairwise-separated-time formulation. This is convergence along *every* such layout, not a density-one or Cesaro substitute. No uniform rate over all bounded observables or all k is asserted.

The pinned manuscript M280 [1], hereafter the *source*, asserts (1.1) for all finite k from ordinary mixing, without an initial standardness assumption. The local applicability catalogue assigns M280 level 5: TR-item-313 [2] studies this very problem, its off-diagonal laws, and concrete sufficient mechanisms. This is a rating of *applicability*, not proof correctness or a claim that the Ram corpus already proves the universal assertion.

Three types of statement must be separated.

1. **Source claims and dependencies.** The saturated array, operator-square, and measurable finite-block construction is the source's route to its universal theorem. Its hypotheses cannot be replaced by marginal independence or an arbitrary joining.

2. **Local results proved here.** The information estimates, projection budgets, Bessel bound, signed residual inequality, and approximation extensions below have explicit hypotheses and proofs. They do not require the global claim.

3. **Conditional consequences.** The positive-integral contradiction and the extensions to endomorphisms and flows follow if the source's structural chain is correct. The homoclinic and Gaussian routes require additional premises not deduced here from ordinary mixing.

A bounded review and a separate local audit found no demonstrated counterexample or actionable circularity. Full reading and that negative finding are not full validation of a difficult new proof.

2 Primary Literature and the Exact Ram Basis

The following comparison was checked against public primary sources as of October 7, 2026. Earlier open-problem statements are dated context; they neither validate nor refute a later manuscript.

Ryzhikov's November 2024 survey [5], introduction and Section 2, presents the unrestricted problem as open at that date and gives an ergodic homoclinic-group sufficient condition (Theorem 2.1). De la Rue [6], Sections 3.1–3.2, explains joining-limit and zero-entropy reductions and why a one-dimensional analogue of the elementary three-dot construction does not settle the question. These sources help distinguish actual limits from merely admissible-looking joinings.

Several results address special classes or weaker quantifiers. Host's publisher abstract [7] states the singular-spectrum joining theorem and its all-order consequence for mixing systems in that class; the full proof was not accessed. Bergelson–Zelada [8], Theorem 1.11 and Corollary 1.12, obtain an almost-all-order statement expressed through large sets of good layouts. This does not replace the every-layout quantifier in (1.1). Kanigowski–Ravotti [9], Theorem 6.8, prove all-order mixing for mixing flows satisfying their locally uniformly shearing hypothesis. Only the relevant statement and surrounding mechanism, not that entire long paper, were reviewed.

Two 2026 preprints add recent context without independently supporting M280's universal conclusion. Ryzhikov [10] revisits the rank-one/Kalikow setting. Wang [11] studies semisimple partially hyperbolic systems under a strong spectral-gap hypothesis, including effective multiple mixing. Their abstracts and relevant opening hypotheses were checked; their full proofs were not audited. A bounded literature search did not establish an independently verified universal resolution against which M280 could simply be checked.

Austin's extension machinery [12], Theorem 3.11 and Lemma 3.12, provides an energy-increment precedent, not a theorem about M280's custom admissible array category. Green's dissociated-set notes [13], Section 2.2, Lemmas 14–15 and Proposition 17, provide a classical context for exponential-moment estimates. No historical novelty is claimed for convexity, entropy duality, Bessel's inequality, or their standard functional-analytic ingredients. The contribution here is their explicit application, strengthening, and error accounting at the source's steps.

2.1 Exact local texts

The mathematical body of TR-item-313 was read, not just a corpus map. Its sections on actual off-diagonal laws, marginal obstructions, quantitative homoclinic residuals, physical Gram optimization, and Gaussian covariance uncertainty are used below. Its historical open-problem discussion must be read with its own date in mind.

Two supporting Ram chapters were read in full: Volume 21, *Physical Metrics and Full Residuals* [3], especially the physical Gram, orthogonal residual decomposition, and approximate-metric statements; and Volume 21, *Extremal Geometry* [4], especially finite moment duality and uncertainty budgets. Their concrete roles are the norm of an actual averaged Koopman image and its omitted physical tail, and the full joint-law constraints in a finite moment problem. Coordinate changes alone do not establish an ergodic closure theorem.

3 The Standard Zero-Entropy Witness

Source Section 2 (`prop:zero-entropy-witness`) makes a delicate reduction. Suppose $k \geq 3$ is the least failing order. After passing to a subsequence, fixed witness sets and diverging-gap layouts have a discrepancy of one sign bounded away from zero. Let α be their finite partition, and write $\alpha_i = T^{-i}\alpha$. Work in the completed sigma-field generated by all integer translates of α, and put

$$\mathcal{F}_{-m} = \sigma(\alpha_i : i \leq -m), \qquad \mathcal{C} = \bigcap_{m \geq 1} \mathcal{F}_{-m}.$$

All sigma-fields and equalities are taken modulo null sets. The remote past $\mathcal{C}$ is invariant in both time directions.

3.1 The order of tail replacement

Put $\bar{g}_i = \mathbb{E}(g_i \mid \mathcal{C})$ for the witness indicators. Replace factors in descending observation-time order. For factor i, translate its time to zero. All earlier factors are measurable in $\mathcal{F}_{-g}$, where g is the preceding gap; all already replaced later factors lie in $\mathcal{C}$. Their product is bounded by one. The error is at most

$$\|\mathbb{E}(g_i \mid \mathcal{F}_{-g}) - \mathbb{E}(g_i \mid \mathcal{C})\|_1 \longrightarrow 0 \tag{3.1}$$

by reverse martingale convergence. The first factor is replaced exactly, because the remaining product is already $\mathcal{C}$-measurable. Conditioning arbitrary unmodified later factors this way would be invalid.

3.2 The zero-entropy assertion

Let β be a finite $\mathcal{C}$-measurable partition. For entropy rate h, finite-window approximation gives

$$h(\alpha \vee \beta) \le h(\alpha) + H\left(\beta \,\middle|\, \bigvee_{|i|\le M} \alpha_i\right) \longrightarrow h(\alpha).$$

Thus $h(\alpha \vee \beta) = h(\alpha)$. Every translate of β lies in the remote α-past, hence in the full α-past before any specified time. The chain rule and monotonicity under additional conditioning give

$$H(\alpha_1, \dots, \alpha_n \mid \beta_1, \dots, \beta_n) \ge nH(\alpha_0 \mid \sigma(\alpha_i : i < 0)) = nh(\alpha).$$

Divide the joint entropy identity by n and pass to its rate. It follows that $h(\beta) = 0$. The word "tail" is not a substitute for this argument.

Approximate the $\bar{g}_i$ by fixed finite-valued $[0,1]$-valued functions on one finite tail partition β. The correlation discrepancy changes by at most twice the sum of their L^1 errors. Choose these errors before taking large layouts. The resulting two-sided finite-symbol law is standard, has zero entropy, and inherits all lower mixing orders. Centering removes proper subproducts by those lower-order hypotheses; changing one sign gives a positive centered witness.

No regular conditional law on the arbitrary original space is used. The standard object is the finite-symbol pushforward law and its completed measure algebra, not the original point space. Later disintegrations are on this factor and the source's standard extensions.

4 The Ordered Array Contract

Source Section 3 begins with laws of actual orbit samples at times $\sum_{\ell=1}^{n} d_{j\ell}(b_\ell)$. It takes successive ultralimits, with the last array axis first. Holding the other indices fixed, mixing of orders below k gives independence of every proper family of last-axis planes; later limits preserve this finite product. Fixing an axis at one slot removes a common time translation. Neither interchange of ultralimits nor exchangeability of axes follows.

An admissible array object has local state Y, local parameter U, maps to X, U, and compatible laws at every array level. The laws project to the original ordered arrays, are consistent on every fixed-axis face, and retain proper last-plane independence. *Conditional on the entire parameter array*, every proper family of last planes is independent, and each plane's conditional law depends only on its own plane parameters. The local X and U coordinates are independent. Extensions

preserve the whole old laws and the old parameter as a factor of the new one. The extensions themselves are not assumed mixing.

4.1 Saturation and full conditioning

Saturation maximizes conditional-expectation energies over this exact extension class using countably many dense tests and successive near-maximizers. Inverse-limit closure is a proof obligation: finite-stage cylinder identities must pass through martingale convergence. A further extension of the limit is a competitor over each earlier stage. Thus equality of nested projection norms gives equality of projections, without requiring an unattained supremum to be attained.

Rolling an axis into the local state adds parent coordinates to the parameter. Extra conditioning can destroy independence. The source uses its local-conditioning lemma and parent-screening identity, not unconditional independence alone. The induction on the number of parents uses saturation of an already established smaller-parent extension. At most $k-2$ parents, together with a separate projection slot, keeps the relevant families proper.

Square locality and grouped missing-cell locality use the *whole* parameter array. Row laws are derived by rolling, not by swapping axes. For parameter geometry, data containing outside-plane states are needed to reverse the screening kernel. This supplies conditionally iid row and column tuples over a shared local parameter, essential in the later label descent.

4.2 Operator and label dependencies

Table 1 summarizes the required source chain. These proofs were read, but the infinite construction was not independently rebuilt. No finite Gram calculation below certifies this table.

Table 1: Dependencies retained from source Sections 3–5.

Step	Required structure	Invalid shortcut
Saturation	All array levels, complete data sigma-fields, inverse-limit closure within the admissible class.	Arbitrary couplings or only marginal parameter conditioning.
Square locality	Parent screening, own-plane kernels, all relevant nonempty grouped row and column splits.	Extra conditioning assumed harmless, or axes assumed exchangeable.
Operator square	Both $L_A J_t = J_s L_B$ and $L_A^* J_s = J_t L_B^*$; restricted active and adjoint range spans.	The same polar conclusions for arbitrary contractions.
Common exact lines	Commuting active projections, essential range density, polar group, contraction completeness.	An active space assumed to be an algebra or exact-line span.
Measurable labels	Countable orthogonal lines, spectral enumeration, conditional iid geometry and support.	Pointwise eigenvector choices assumed measurable.
Finite blocks	Explicit counting kernels, partial-unitary identities, positive L^2 tests, product parameter marginals.	Normalization of infinite counting mass, or countable sets used in place of finite blocks.

Under independent proper marginals and $k \geq 3$, exact unimodular tuples whose product is constant have unique partners; distinct exact lines are orthogonal. Separability gives countability. In the operator square, fixed positive squares can be injective without a gap. From $A^2N = NB^2$, functional calculus gives $AN = NB$. A bounded relation $DB = 0$ implies $D = 0$ because B has dense range. A bounded B^{-1} is neither needed nor established.

For the polar group's simultaneous diagonalization, averaging a positive injective trace-class operator under its invariant probability gives a positive injective trace-class operator commuting with the group. Its finite-dimensional positive eigenspaces span the space and can be simultaneously diagonalized. This is not a compact-resolvent assertion about an unrelated operator.

For active projections $\Pi(p), \Pi(q)$ that are conditionally iid, put $B = \mathbb{E}_q \Pi(q)$. A countable dense family shows $\Pi(p) \operatorname{Ker} B = 0$ almost surely. An active vector orthogonal to every common exact line has $\Pi(q)v = 0$, hence belongs to $\operatorname{Ker} B$, and must vanish. Later column-membership variables are genuinely 0-1; conditional independence and matching give $r = r^2$, and the support argument forces $r > 0$. Averaged projections alone would not give this discrete conclusion.

The lifted measure is counting measure in each label fibre integrated against the parameter law. It is sigma-finite, generally not a probability. Unique matching provides probability transition kernels without tilting the parameter law by label multiplicity. The identities $M = M^*MM = MMM^*$ make M unitary on its common range and zero on its orthogonal complement. A strictly positive L^2 test and its countable orbit vector yield countably many levels; positivity bounds each level's mass, so its fibre sections are finite. These finite sections, not arbitrary countable sums, enter the final bounded-convergence argument.

5 Quantitative Screening and Its Stability Boundary

The exact saturation argument admits a quantitative local certificate; it does not quantitatively construct the extensions.

Proposition 5.1 (Projection-energy leakage)**.** *Let $\mathcal{D} \subset \mathcal{D}'$ be sigma-fields on one probability space, with conditional-expectation projections P, Q on L^2. For $F \in L^2$, set $\eta = \|QF\|_2^2 - \|PF\|_2^2$. Then*

$$\eta = \|QF - PF\|_2^2 \geq 0, \qquad |\langle QF - PF, G\rangle| \leq \sqrt{\eta}\, \|G\|_2 .$$

If G is $\mathcal{D}'$-measurable, the same bound holds with $QF - PF$ replaced by $F - PF$.

Proof. Nested orthogonal projections satisfy $PQ = QP = P$. Thus $QF = PF + (QF - PF)$ is an orthogonal decomposition. Apply Cauchy–Schwarz. In the last assertion, $F - QF$ is orthogonal to G. □

For $|G| \leq 1$, the bound is $\sqrt{\eta}$. If only the *norm* gain is at most ε, then $\eta \leq 2\|F\|_2 \varepsilon$, not generally ε^2. The fields must still be the source's full data fields; approximate marginal identities do not certify complete screening.

Proposition 5.2 (Dense-range cancellation with a budget)**.** *Let B be a positive contraction on a Hilbert space and let D be bounded with $\|DB\| \leq \varepsilon$. For $0 < \gamma \leq 1$, let $E_\gamma = \mathbf{1}_{[\gamma,1]}(B)$. Then*

$$\|DE_\gamma\| \leq \varepsilon/\gamma, \qquad \|Dv\| \leq (\varepsilon/\gamma)\|v\| + \|D\|\, \|(I - E_\gamma)v\| .$$

Proof. Bounded functional calculus gives $\|B^{-1}E_\gamma\| \leq 1/\gamma$. Since $DE_\gamma = DB(B^{-1}E_\gamma)$, the first assertion follows. Split v using E_γ. □

For example, $B = \operatorname{diag}(1, \varepsilon)$ and $D = \operatorname{diag}(0, 1)$ give $\|DB\| = \varepsilon$ but $\|D\| = 1$. Exact dense-range cancellation can be sound while its approximate analogue is ill-conditioned. Approximate locality does not produce exact active lines without a gap or a controlled small-spectrum tail. This limits a quantitative version, not the source's exact argument.

6 Sharp Finite-Word Entropy and Information Bounds

The following is a self-contained strengthening of source Section 6's dissociated-character cancellation. All logarithms are natural.

Let Γ be a countable abelian group and let χ_ℓ be measurable functions on a probability space (Ω, η) satisfying

$$|\chi_\ell| = 1, \quad \chi_0 = 1, \quad \chi_\ell \chi_{\ell'} \in \mathbb{T}\chi_{\ell+\ell'}, \quad \mathbb{E}_\eta \chi_\ell = 0 \quad (\ell \neq 0). \tag{6.1}$$

The multipliers in the third condition are *constant scalar phases*. A list $\ell_1, \dots, \ell_n$ is dissociated if $\sum_m \epsilon_m \ell_m \neq 0$ for every nonzero $\epsilon \in \{-1, 0, 1\}^n$. Repetitions of an index in that sum are not allowed.

Lemma 6.1 (All-parameter moment bound)**.** *For every deterministic $w \in \overline{\mathbb{D}}^n$, where $\overline{\mathbb{D}} = \{z \in \mathbb{C} : |z| \leq 1\}$, and every $\theta > 0$,*

$$\mathbb{E}_\eta \exp\left(\theta \sum_{m=1}^{n} \Re(w_m \overline{\chi_{\ell_m}})\right) \leq (\cosh \theta)^n \leq e^{n\theta^2/2}.$$

Proof. Convexity on $[-1, 1]$ gives $e^{\theta x} \leq \cosh \theta + x \sinh \theta$. The right side is positive, so multiply these inequalities for $x_m = \Re(w_m \overline{\chi_{\ell_m}})$. Every nonconstant term in the expansion is a linear combination of signed products of distinct characters. By projectivity it is a scalar multiple of a character with nonzero label, so its expectation vanishes. Only $(\cosh \theta)^n$ remains. Finally, $\log \cosh \theta = \int_0^\theta \tanh t \, \mathrm{d}t \leq \theta^2/2$. □

There is no small-θ restriction. Pairwise orthogonality is not enough: for independent fair signs X, Y, the three functions X, Y, XY are pairwise orthogonal, but the exponential expansion has an additional $(\sinh \theta)^3$ term since their product is one.

Theorem 6.2 (Sharp information curve)**.** *Assume (6.1) and dissociation. Let $W = (F_1, \dots, F_n)$ be any measurable $\overline{\mathbb{D}}^n$-valued vector, arbitrarily coupled to $C = (\chi_{\ell_1}, \dots, \chi_{\ell_n})$, and put*

$$A = \frac{1}{n} \sum_{m=1}^{n} |\mathbb{E}_\eta F_m \overline{\chi_{\ell_m}}| \in [0, 1].$$

For $Q = \operatorname{Law}(W, C)$, $P = \operatorname{Law}(W) \otimes \operatorname{Law}(C)$, assume $I(W; C) = D(Q\|P) < \infty$. Then

$$J(A) \leq \frac{I(W; C)}{n}, \qquad J(a) = \frac{(1+a)\log(1+a) + (1-a)\log(1-a)}{2}. \tag{6.2}$$

Here $0 \log 0 = 0$, $J(0) = 0$, and $J(1) = \log 2$. Consequently,

$$A \leq J^{-1}(\min\{I(W; C)/n, \log 2\}) \leq \min\{1, \sqrt{2I(W; C)/n}\}. \tag{6.3}$$

If W is finite valued, then $I(W; C) \leq H(W) < \infty$, so I may be replaced by $H(W)$ in both bounds.

Proof. Choose deterministic phases ω_m aligning the means: $\omega_m \mathbb{E}(F_m \overline{\chi_{\ell_m}}) = |\mathbb{E}(F_m \overline{\chi_{\ell_m}})|$. Set $S(w,c) = \sum_m \Re(\omega_m w_m \bar{c}_m)$. For fixed w, Lemma 6.1 applies. Integrating over the *independent reference law* P gives $\mathbb{E}_P e^{\theta S} \le (\cosh\theta)^n$. Independence is not asserted under the actual law Q.

For a bounded S, let $\mathrm{d}P_\theta = e^{\theta S}\,\mathrm{d}P / \mathbb{E}_P e^{\theta S}$. Nonnegativity of $D(Q\|P_\theta)$ gives

$$\theta n A = \theta \mathbb{E}_Q S \le D(Q\|P) + \log \mathbb{E}_P e^{\theta S} \le I(W;C) + n \log\cosh\theta.$$

The supremum of $\theta a - \log\cosh\theta$, over $\theta > 0$, is $J(a)$: for $0 < a < 1$, the optimizer is $\theta = \operatorname{artanh}(a)$; the endpoints follow by limits. Moreover $J'(a) = \operatorname{artanh}(a) > 0$ and $J''(a) = 1/(1-a^2) \ge 1$, so $J(a) \ge a^2/2$. This proves (6.2)–(6.3).

For finite W, the conditional probabilities given C exist on these standard Borel vector spaces and $I(W;C) = H(W) - \mathbb{E}H(W \mid C) \le H(W)$. No standardness of a larger ambient sample space is required, since the argument uses the vector laws. □

For completeness, the finite-word entropy version also has an elementary proof that avoids mutual-information notation. For a word w of probability $p_w > 0$, the conditional law $\eta_w = \eta(\,\cdot \mid W = w)$ has $D(\eta_w\|\eta) = \log(1/p_w)$. Jensen and restriction of a positive integral give

$$\theta \mathbb{E}_{\eta_w} S_w \le \log \mathbb{E}_{\eta_w} e^{\theta S_w} \le \log(1/p_w) + n \log\cosh\theta, \quad S_w = \sum_m \Re(\omega_m w_m \overline{\chi_{\ell_m}}).$$

Average over w. The mean on the left is $\theta n A$, yielding $J(A) \le H(W)/n$, and in particular

$$A \le \min\{1, \sqrt{2H(W)/n}\}. \tag{6.4}$$

When $H(W) = 0$, take $\theta \downarrow 0$ to get $A = 0$. There is no claim that character orthogonality persists after conditioning on the word. The original unconditional character law is the comparison law throughout.

6.1 Sharpness and the scope of the improvement

Take independent fair Rademacher characters C_m of the standard basis of $(\mathbb{Z}/2\mathbb{Z})^n$. The basis list is dissociated in the distinct-index sense above. Independently take noise signs Z_m with $\Pr(Z_m = -1) = \varepsilon \in [0, 1/2]$, and set $W_m = C_m Z_m$. Then

$$A = 1 - 2\varepsilon, \qquad I(W;C) = n\{\log 2 - h_b(\varepsilon)\} = nJ(A),$$

where h_b is binary entropy with natural logarithms. For $0 < A < 1$, the actual density relative to the product law is

$$\prod_{m=1}^{n} (1 + A w_m c_m) = \frac{\exp\{\operatorname{artanh}(A) \sum_m w_m c_m\}}{\cosh(\operatorname{artanh}(A))^n}.$$

Thus the entropy inequality is attained. The endpoints follow directly or by limits. Sharpness concerns the information form, not equality in the weaker square-root or word-entropy bound for every parameter.

The source uses a small-parameter bound of the form $e^{2n\theta^2}$. The all-parameter cosh estimate and direct entropy transfer remove the likely-word truncation and give an explicit finite-n bound. For source words with $H_n = o(n)$, cancellation follows immediately. If an independent estimate gives $H_n \le Cn^\alpha$, $\alpha < 1$, then (6.4) gives $\sqrt{2C}\, n^{-(1-\alpha)/2}$. This is a rate for this dissociated block, not for all polynomial orbits, all layouts, or the full mixing theorem. Neither entropy nor mutual information is declared computable from finitely many moments.

6.2 A defect-aware variant

Let R be a reference law of C, and suppose uniformly for all deterministic $|w_m| \le 1$ and all $\theta > 0$,

$$\log \mathbb{E}_R e^{\theta S_w} \le \tfrac{1}{2} v_n \theta^2 + d_n, \qquad v_n > 0, \quad d_n \ge 0.$$

For the actual joint law Q, if $\mathcal{I} = D(Q \| \operatorname{Law}(W) \otimes R) < \infty$, the same entropy proof and optimization give

$$A \le \min\left\{1, \frac{\sqrt{2v_n(\mathcal{I} + d_n)}}{n}\right\}. \tag{6.5}$$

The zero numerator case follows by $\theta \downarrow 0$.

For example, suppose every nonempty signed product of distinct characters has expectation of modulus at most ε under R. The same chord expansion, including its coefficient absolute sums, gives

$$\mathbb{E}_R e^{\theta S_w} \le (\cosh \theta)^n \left[1 + \varepsilon\{(1 + \tanh \theta)^n - 1\}\right].$$

Thus $v_n = n$ and $d_n = \log(1 + \varepsilon(2^n - 1))$ are valid. The exponential-in-n cost is explicit. A few pairwise tests cannot be silently substituted for all these higher signed-product conditions.

7 Orbit Cancellation Beyond One-Site Observables

We retain the exact setting of source Section 6. Let T be a mixing, zero-entropy, two-sided finite-alphabet process on (X, μ). Let η be any probability coupling carrying $x \in X$ with marginal μ and characters (6.1). Let G be an automorphism of Γ. No invariance of η under a time action is assumed.

7.1 The algebraic orbit alternatives

The source's algebraic reduction has two branches. If the orbit of ℓ is dissociated, the entropy estimate applies. Otherwise a finite relation $\sum_{j=a}^{b} \epsilon_j G^j \ell = 0$, with $\epsilon_a, \epsilon_b \in \{-1, 1\}$, makes its two-sided cyclic module finitely generated: both forward and backward powers reduce to a finite window. Its torsion subgroup is finite, and the induced automorphism of the free quotient is an integral lattice automorphism.

If an eigenvalue λ has $|\lambda| > 1$, a left eigenfunctional detects the cyclic generator. Choose p with $|\lambda|^p > 2$. On each residue progression, a signed sum is nonzero because the largest term in that eigenfunctional dominates the sum of all previous terms. Each progression is therefore dissociated.

Otherwise all eigenvalues have modulus one, since the determinant has modulus one. The monic integral characteristic polynomials of powers have bounded coefficients, so there are only finitely many such polynomials. Every power of each eigenvalue lies in the finite union of their roots; hence all eigenvalues are roots of unity. Take a power p unipotent on the free quotient and the identity on the finite torsion subgroup. For $N = G^p - I$, some N^d maps the module into its torsion subgroup, and $N^{d+1} = 0$. Thus on each residue class

$$G^{pm+j}\ell = \sum_{r=0}^{d} \binom{m}{r} N^r G^j \ell.$$

This is a group-valued polynomial in the binomial basis. No invariant splitting of free and torsion parts is needed. The progression length and degree depend on the orbit; no uniform effective bound is asserted.

7.2 Polynomial cancellation with arbitrary coupling

For a group polynomial $P(m)$, fixed $p \geq 1$, and bounded centered F, define

$$c_N(F,P) = \frac{1}{N}\sum_{m=0}^{N-1}\left|\int F(T^{pm}x)\overline{\chi_{P(m)}}\,\mathrm{d}\eta\right|.$$

If P is constant, condition its character on x. This yields a single fixed bounded test on X, so ordinary mixing and $\mu F = 0$ give $c_N \to 0$.

For higher degree put

$$F_h = (F\circ T^{ph})\bar{F}, \qquad \rho_h = \mu(F_h), \qquad \Delta_h P(m) = P(m+h) - P(m).$$

Projective scalar phases do not affect absolute values. The h-lag inner products in the fixed Hilbert space $L^2(\eta)$ reduce to correlations of F_h with $\chi_{\Delta_h P(m)}$. Subtract ρ_h, apply the lower-degree assertion to $F_h - \rho_h$, and bound the leftover by $|\rho_h|$. Mixing gives $\rho_h \to 0$. The following finite estimate makes the order of limits precise.

Proposition 7.1 (Finite van der Corput budget)**.** *If $\|F\|_\infty \leq M$, then for $1 \leq H \leq N$,*

$$c_N(F,P)^2 \leq \left(1+\frac{H-1}{N}\right)\left[\frac{M^2}{H} + \frac{2}{H}\sum_{h=1}^{H-1}\left(1-\frac{h}{H}\right)\left\{c_{N-h}(F_h-\rho_h,\Delta_h P) + |\rho_h|\right\}\right]. \tag{7.1}$$

Proof. Choose phases aligning the scalar means and set $v_m = \omega_m F(T^{pm}x)\overline{\chi_{P(m)}}$. Then $c_N \leq \|N^{-1}\sum_m v_m\|_{L^2(\eta)}$. Extend $v_m = 0$ outside $0 \leq m < N$. The identity

$$H\sum_{m=0}^{N-1} v_m = \sum_{r=-H+1}^{N-1}\sum_{a=0}^{H-1} v_{r+a}$$

and Cauchy–Schwarz bound the squared norm by $N+H-1$ times the sum of the squared window norms, divided by H^2N^2. Expanding those norms gives the diagonal bound HNM^2 and off-diagonal terms $2\sum_{h=1}^{H-1}(H-h)\sum_{m<N-h}|\langle v_m, v_{m+h}\rangle|$. Each inner product is bounded by the centered lower-degree correlation plus $|\rho_h|$. Drop the harmless factor $(N-h)/N \leq 1$ to obtain (7.1). □

First let $N \to \infty$ with H fixed; then let $H \to \infty$. This proves the induction. Reversing these operations or asserting a uniform bound for a growing family of mixing tests would require additional information. The proof uses only the fixed X-marginal of η, not fibre invariance.

Theorem 7.2 (Centered L^1 extension)**.** *Under the setting above, every centered $f \in L^1(\mu)$ and every $\ell \in \Gamma$ satisfy*

$$\frac{1}{N}\sum_{r=0}^{N-1}\left|\int f(T^r x)\overline{\chi_{G^r\ell}}\,\mathrm{d}\eta\right| \longrightarrow 0. \tag{7.2}$$

Proof. First let f be a bounded finite-valued finite-cylinder function. The word $(f(T^{pm+j}x))_{m<N}$ is determined by a base-symbol block of length $pN + O(1)$. Zero entropy therefore gives word entropy $o(N)$ for every fixed p, j. After scaling f, Theorem 6.2 proves cancellation on the dissociated progressions. On the polynomial progressions the preceding induction applies to the centered observable $f \circ T^j$. The orbit alternatives cover finitely many residue classes, giving (7.2).

For general centered $f \in L^1$, take its conditional expectations on increasing finite coordinate windows. They are bounded finite-valued centered functions g and converge to f in L^1. For every r and every label,

$$\left| \int (f-g)(T^r x)\overline{\chi_\ell}\, \mathrm{d}\eta \right| \le \|f-g\|_{L^1(\mu)}.$$

Take the Cesaro limsup, and then let the approximation error go to zero. □

The same approximation extends the polynomial-cancellation lemma to centered L^1 observables in any mixing system; zero entropy is needed for the dissociated branch of the combined orbit theorem. The extension neither removes this premise nor proves that the required characters and blocks exist in a general mixing system.

8 Finite Fibres, Positive Integrals, and Bounded Convergence

We now state explicitly what source Sections 5–7 must supply. For almost every parameter u, a fibre law η_u has X-marginal exactly μ, a projective orthogonal character family, a group automorphism G, and a finite nonempty label block B. For a centered finite-valued witness f, the intrinsic block coefficient sum has the representation

$$s_r(u) = \sum_{\ell \in B} \left| \int f(T^r x)\overline{\chi^u_{G^r \ell}}\, \mathrm{d}\eta_u \right|.$$

The expression is intrinsically measurable even if the pointwise identifications of label groups across fibres are not chosen measurably. The source's transport identity yields

$$\int s_r(u)\, \mathrm{d}\nu(u) = c > 0 \quad \text{for every } r. \tag{8.1}$$

One positive coefficient at one time is not enough for (8.1); finite-block transport and the measure-preserving parameter action are indispensable.

Proposition 8.1 (Bessel fibre majorant)**.** *For every such typical fibre and time,*

$$s_r(u) \le \sqrt{|B|}\ \|f\|_{L^2(\mu)}.$$

For merely $f \in L^1$, the bound $s_r(u) \le |B|\,\|f\|_{L^1(\mu)}$ remains valid.

Proof. The labels $G^r\ell$, $\ell \in B$, are distinct; their characters are orthonormal under η_u. Cauchy–Schwarz on the finite coefficient vector followed by Bessel gives

$$s_r(u) \le \sqrt{|B|} \left(\sum_{\ell \in B} \left| \int f(T^r x)\overline{\chi^u_{G^r \ell}}\, \mathrm{d}\eta_u \right|^2 \right)^{1/2} \le \sqrt{|B|}\, \|f(T^r x)\|_{L^2(\eta_u)}.$$

The last norm is $\|f\|_{L^2(\mu)}$ because the X-marginal is μ and T preserves μ. This does not assert invariance of η_u. The L^1 bound follows coefficient by coefficient. □

This improves the source's $|B|\,\|f\|_\infty$ majorant. On each typical fibre, Theorem 7.2, summed over the finite B, gives $A_N(u) = N^{-1}\sum_{r<N} s_r(u) \to 0$. The intrinsic sums are measurable and

uniformly bounded, so bounded convergence gives $\int A_N \to 0$, contradicting (8.1). No integration of a nonmeasurable group identification or interchange with an infinite label sum is needed.

This contradiction is a sound *conditional* last step once the source's structural hypotheses and transport identity are supplied. The use of an auxiliary Cesaro average does not weaken the theorem to Cesaro mixing: the positive witness was obtained from an arbitrary sequence violating the full every-layout statement.

There is also a useful finite error budget. If $0 \le A_N \le M$, $A_N \le e_N$ outside a measurable exceptional set of probability at most β_N, and $|\int A_N - c| \le r_N$, then

$$c \le r_N + e_N + M\beta_N. \tag{8.2}$$

Indeed integrate on the good set and its complement. Pointwise cancellation alone provides no effective values of e_N, β_N, r_N; the formula records what a quantitative construction would have to certify.

9 Actual Off-Diagonal Laws and Marginal Obstructions

TR-item-313 formulates the central object as the actual law

$$\lambda_{\mathbf{t}} = (x, T^{t_1}x, \dots, T^{t_{k-1}}x)_*\mu.$$

On a finite-symbol factor, subsequential limits exist in the compact space of probability laws on the product shift. They are stationary self-joinings. If all lower mixing orders hold and the gaps diverge, their proper marginals are product. The desired conclusion is that *these actual limits* are the full product law.

The following exact example is a necessary negative test for any proposed moment closure:

$$p_\theta(x) = 2^{-k}\left(1 + \theta\prod_{j=0}^{k-1} x_j\right), \quad x \in \{-1,1\}^k, \quad |\theta| \le 1. \tag{9.1}$$

It is nonnegative and normalized. Every proper marginal is uniform product because summing a missing sign kills the parity term, but

$$\mathbb{E}_{p_\theta}\prod_j x_j = \theta, \qquad \|p_\theta - p_0\|_{\mathrm{TV}} = |\theta|/2.$$

Thus even all proper moments do not determine the full joint moment.

For $k = 3$, let two independent two-sided fair-sign processes be $A = (A_n)$, $B = (B_n)$, and put $C_n = A_nB_n$. Each process is Bernoulli and any pair of complete processes is independent, but $A_nB_nC_n = 1$. This is a stationary nonproduct joining with product proper marginals. It is not a nonproduct separated off-diagonal limit of the Bernoulli shift: finite cylinders at sufficiently separated times already factor. It refutes marginal-only closure, not ordinary mixing implying higher-order mixing.

9.1 What finite Ram moment coordinates can certify

For a finite alphabet, the full tensor basis, or Walsh basis in the sign case, is an exact invertible coordinate representation of the joint law. It preserves the distinction just exhibited; discarding the

highest interaction loses the obstruction. Under linear constraints $Ap = b$, $p \geq 0$, a finite extremal test has the primal and dual forms

$$\sup\{c^{\mathsf{T}} p : Ap = b,\ p \geq 0\}, \qquad \inf\{b^{\mathsf{T}} y : A^{\mathsf{T}} y \geq c\}.$$

When the feasible set is nonempty and the objective is bounded, ordinary finite linear-programming duality gives equality. Even without invoking equality, each dual feasible y is a valid upper certificate. This is the concrete role of the extremal Ram framework [4].

If measurements satisfy $\|Ap - b\|_\infty \leq \varepsilon$, the same certificate gives $c^{\mathsf{T}} p \leq b^{\mathsf{T}} y + \varepsilon \|y\|_1$. Other uncertainty sets contribute their support functions. Positivity, marginal, stationarity, and finite consistency constraints can improve a relaxation, but do not establish that every feasible point is a realizable dynamical limit, nor that the saturated operator-square hypotheses hold. Such additional structure must be proved.

10 Signed Homoclinic Certificates and Physical Gram Tails

The quantitative homoclinic theorem in TR-item-313 is a different route with stronger premises. We first prove its finite, observable-specific form and a signed-weight extension. No mixing is needed for this finite inequality.

Let $Uf = f \circ T$, and let $V_a f = f \circ S_a$, where T, S_a are invertible probability-preserving transformations of the same space. Assume $|f_j| \leq 1$. For $0 = t_0 < t_1 < \cdots < t_{k-1}$, put

$$\Delta_k(\mathbf{t}; \mathbf{f}) = \int \prod_{j=0}^{k-1} U^{t_j} f_j \, \mathrm{d}\mu - \prod_{j=0}^{k-1} \mu(f_j), \quad e_{aj}(t) = \left\| U^{-t} V_a^{-1} U^t f_j - f_j \right\|_1 .$$

Let $z_1, \ldots, z_m \in \mathbb{C}$ satisfy $\sum_a z_a = 1$, and define

$$a_z = \left\| \sum_a z_a V_a f_0 - \mu(f_0) \right\|_2 ,$$

$$b_z(\mathbf{t}) = \sum_a |z_a| \sum_{j=1}^{k-1} e_{aj}(t_j), \qquad \Lambda = \sum_a |z_a|.$$

Theorem 10.1 (Signed finite residual inequality)**.** *For these data,*

$$|\Delta_k(\mathbf{t}; \mathbf{f})| \leq a_z + b_z(\mathbf{t}) + |\mu(f_0)|\, |\Delta_{k-1}((0, t_2 - t_1, \ldots, t_{k-1} - t_1); (f_1, \ldots, f_{k-1}))|. \tag{10.1}$$

Proof. Invariance under S_a gives

$$\int f_0 \prod_{j \geq 1} U^{t_j} f_j \, \mathrm{d}\mu = \int V_a f_0 \prod_{j \geq 1} V_a U^{t_j} f_j \, \mathrm{d}\mu.$$

Replace the latter shifted factors one at a time by $U^{t_j} f_j$. All other factors are bounded by one, so the error is at most

$$\sum_{j \geq 1} \left\| V_a U^{t_j} f_j - U^{t_j} f_j \right\|_1 = \sum_{j \geq 1} e_{aj}(t_j).$$

Multiply by z_a, sum, and use $\sum_a z_a = 1$. The accumulated error is at most b_z. Replace $\sum_a z_a V_a f_0$ by its mean $\mu(f_0)$; Cauchy–Schwarz and the bounded product give error at most a_z. The remaining $(k-1)$-correlation is shifted by t_1 using T-invariance. Subtract the product of means. □

For nonnegative real weights the theorem is TR-item-313's convex residual mechanism. Complex affine weights are allowed, but their total variation Λ amplifies error; no positivity benefit is then available.

10.1 Physical Gram optimization

Let $h = f_0 - \mu(f_0)$, $v_a = V_a h$, and use the inner product $\langle u, v\rangle = \int \bar{u}v \, \mathrm{d}\mu$, linear in its second argument. For the physical Gram matrix $G_{ab} = \langle v_a, v_b\rangle$,

$$a_z^2 = z^* G z.$$

If d_a bounds $\sum_{j\geq 1} e_{aj}(t_j)$, then minimizing

$$\sqrt{z^* G z} + \sum_a d_a |z_a|, \qquad \sum_a z_a = 1, \tag{10.2}$$

is a convex optimization problem over real or complex affine weights. More importantly, *any* feasible weights give a certificate; no unverified claim about reaching the optimum is required.

A matrix formed from coordinates is useful only with its physical synthesis metric. For a finite frame $J : \mathbb{C}^r \to L^2$, the metric is J^*J; if the frame is independent, the orthogonal physical projection is $P = J(J^*J)^{-1}J^*$. This is the exact point of the physical-metric Ram source [3]. A Euclidean coefficient residual is not generally the physical residual.

Proposition 10.2 (Orthogonal tail and metric uncertainty)**.** *Let P be an orthogonal projection in L^2, let $G^P_{ab} = \langle Pv_a, Pv_b\rangle$, and suppose $r_a \geq \|(I-P)v_a\|_2$. Then*

$$a_z^2 \leq z^* G^P z + \left(\sum_a |z_a| r_a\right)^2. \tag{10.3}$$

If a Hermitian computed matrix $\widehat{G}$ satisfies $|\widehat{G}_{ab} - G^P_{ab}| \leq \eta$ entrywise, then

$$a_z^2 \leq z^* \widehat{G} z + \eta\Lambda^2 + \left(\sum_a |z_a| r_a\right)^2. \tag{10.4}$$

Proof. Orthogonality gives the exact decomposition

$$\left\|\sum_a z_a v_a\right\|_2^2 = \left\|\sum_a z_a P v_a\right\|_2^2 + \left\|\sum_a z_a (I-P) v_a\right\|_2^2.$$

Bound the second term by the squared triangle bound. The entrywise perturbation satisfies $|z^*(G^P - \widehat{G})z| \leq \eta(\sum_a |z_a|)^2$. □

The certified right side of (10.4) is nonnegative in exact arithmetic. Negative floating-point output is a reason to audit roundoff and bounds, not to discard an unbudgeted negative quantity. Likewise entrywise metric closeness is not a relative spectral bound without a condition number or lower eigenvalue estimate.

If f_j is replaced by an approximation $\widehat{f}_j$, the physical defect satisfies

$$e_{aj}(t) \leq \left\|U^{-t} V_a^{-1} U^t \widehat{f}_j - \widehat{f}_j\right\|_1 + 2\left\|f_j - \widehat{f}_j\right\|_1$$

because both operators are L^1-isometries. Weighted propagation costs $2\Lambda \sum_{j\geq 1} \left\|f_j - \widehat{f}_j\right\|_1$. This tail and approximation accounting is essential for a trustworthy finite-frame certificate.

10.2 A finite example where signed weights help

On the uniform six-cycle, let $h(x) = \cos(\pi x/3)$ and $v_a(x) = h(x+a)$, $a = 0, 1, 2$. Direct trigonometry gives

$$G = \begin{pmatrix} 1/2 & 1/4 & -1/4 \\ 1/4 & 1/2 & 1/4 \\ -1/4 & 1/4 & 1/2 \end{pmatrix}, \qquad z = (1, -1, 1)^{\mathsf{T}}, \qquad Gz = 0, \quad \sum_a z_a = 1.$$

Thus the signed residual is zero and $\Lambda = 3$. For convex weights, projection onto $v_1/\|v_1\|$ is at least $1/(2\sqrt{2})$; this follows from $\langle v_1, v_0\rangle = \langle v_1, v_2\rangle = 1/4$, $\langle v_1, v_1\rangle = 1/2$. The lower bound is attained by $(1/2, 0, 1/2)$, because $(v_0 + v_2)/2 = v_1/2$. Hence the minimum convex residual is $1/(2\sqrt{2})$.

For a common certified cost $d_a = d$, signed weights give $3d$; the best convex certificate is $d + 1/(2\sqrt{2})$. The signed choice improves it when $d < 1/(4\sqrt{2})$. This example proves a finite optimization advantage, not mixing of a finite cyclic system or a counterexample to any mixing theorem.

10.3 What produces qualitative or quantitative all-order mixing

For a homoclinic automorphism S, the defining strong L^2 convergence is

$$U^{-n}V^{-1}U^n f \longrightarrow f \quad (n \to \infty)$$

for every $f \in L^2$. For bounded tests this is equivalent to convergence of the corresponding L^1 defects, by truncation and $\|g\|_2^2 \le \|g\|_\infty \|g\|_1$.

If a group of such automorphisms acts ergodically, the closed convex hull of the orbit of a centered $h \in L^2$ contains zero. Indeed its unique minimum-norm point is fixed by the group, hence constant, and still centered. A finite convex average can therefore make a_z arbitrarily small. Choose that finite family *first*; then let all positive times grow so its finitely many homoclinic defects tend to zero. Induction in (10.1) yields all-order mixing. This is the stronger-premise route in TR-item-313 and Ryzhikov [5]; ordinary mixing has not been shown here to provide that ergodic homoclinic group.

For a finite averaging scheme with certified $a_N \le AN^{-\alpha}$, $\alpha > 0$, and $b_N \le BN\rho(g)$, optimizing its size gives $O(\rho(g)^{\alpha/(1+\alpha)})$, with constants depending on A, B, α and rounding. The estimate is useful only if these premises are independently proved in the particular model. It is not an effective rate for arbitrary ordinary mixing.

11 Gaussian Joint-Law Closure as a Separate Case

A second concrete route in TR-item-313 is available for a genuinely Gaussian process, not for an arbitrary process with the same covariance. Let $X = (X_0, \dots, X_{k-1})$ be a centered finite-dimensional jointly Gaussian vector, partitioned into coordinate blocks. Let Y have the same within-block covariances as X and independent blocks. For smooth bounded functions φ_j with

bounded derivatives, Gaussian interpolation between their covariance matrices gives

$$\left|\mathbb{E}\prod_j \varphi_j(X_j) - \prod_j \mathbb{E}\varphi_j(X_j)\right| \\ \leq \sum_{i<j} \sum_{\substack{a \text{ in block } i \\ b \text{ in block } j}} |\operatorname{Cov}(X_a, X_b)|\, \|\partial_a\varphi_i\|_\infty \|\partial_b\varphi_j\|_\infty \prod_{\ell\neq i,j} \|\varphi_\ell\|_\infty . \tag{11.1}$$

Here derivatives within a block are indexed by that block's coordinates. The covariance interpolation is positive semidefinite because it is a convex combination of two covariance matrices.

For completeness, for positive-definite interpolants with density p_s, differentiation of the Gaussian Fourier transform gives $\partial_s p_s = \frac{1}{2}\sum_{a,b} \dot\Sigma_{ab}\partial_{ab}p_s$. Two integrations by parts transfer derivatives to the product test. The two symmetric cross-block terms cancel the factor $1/2$; the within-block differences vanish. Integrate in s and bound. Adding εI, then taking $\varepsilon \downarrow 0$, handles degenerate covariances by dominated convergence. For indicators or nonsmooth bounded tests one first uses an appropriate L^1 approximation under the fixed marginal Gaussian laws, with an explicit telescoping approximation error.

Entrywise covariance uncertainty adds the same derivative weights times the entrywise error bounds to (11.1). If all inter-block covariances vanish along the separated layouts, fixed finite-cylinder Gaussian laws converge to their product, and cylinder approximation gives the corresponding all-order conclusion. The closure of the *joint* Gaussian family is the essential premise. The parity example in Section 9 explains why covariance coordinates alone cannot transfer this result to general laws.

12 Conditional Endomorphism and Flow Consequences

These are deductions from the source's asserted universal theorem, not independent ways to prove its missing structural premises.

12.1 Endomorphisms on arbitrary probability spaces

For a possibly noninvertible mixing probability-preserving map, fix the finite collection of tested sets and code their forward names. This gives a standard one-sided finite-symbol stationary factor. Its two-sided natural extension is mixing: first verify mixing on finite cylinders by shifting all their indices into the nonnegative range and using mixing of the one-sided factor; then use cylinder approximation. If the source's invertible theorem holds, it applies to this natural extension, and the tested correlations descend to the original space. No natural extension of the entire arbitrary original space is required.

12.2 Strongly continuous mixing flows

Suppose $(S_t)_{t\in\mathbb{R}}$ is a probability-preserving flow, mixing as $|t| \to \infty$, with strongly continuous Koopman action on L^1. Then $T = S_1$ is mixing. Given real layouts, subtract their first time and write $t_i = n_i + \theta_i$, $0 \leq \theta_i < 1$. Along any potentially bad sequence choose a subsequence with $\theta_i \to \theta_i^*$ for every one of the finitely many i. For indicators f_i, strong L^1 continuity gives

$$\left\| f_i \circ S_{\theta_i} - f_i \circ S_{\theta_i^*} \right\|_1 \to 0.$$

Invariance and telescoping bound the product-correlation replacement by the sum of these errors. Integer gaps still diverge, so the source's discrete theorem, if correct, applies to the *fixed* observables $f_i \circ S_{\theta_i^*}$. Every putative bad sequence is contradicted. Bounded functions follow by approximation. Strong continuity, or an equivalent sufficient indicator-continuity hypothesis, cannot simply be dropped on an arbitrary probability space.

13 Advantages, Costs, and Focused Further Work

The direct applicability of the Ram framework is strongest where it preserves an actual probability law, physical norm, or complete constraint set. The source's abstract construction and these quantitative certificates answer different questions.

Advantages. The sharp information curve retains arbitrary coupling and removes an unnecessary small-parameter restriction. It produces a usable finite-word estimate whenever entropy or mutual information is independently controlled. Projection energy converts to a bounded-test screening residual without an unspecified constant. Bessel reduces the finite-block majorant from linear to square-root dependence on block size for L^2 tests. The signed homoclinic program can reduce a true residual while charging for error amplification, and orthogonal tails prevent a compressed calculation from being mistaken for a full-space certificate.

Costs and boundaries. Dissociation is a higher-order condition, not pairwise orthogonality. A finite entropy certificate does not build the source's exact lines. The source's positive operators need not have gaps, so approximate locality may be unstable. Finite-block identification is not automatically measurable, and arbitrary couplings need not be time invariant. Signed weights can be numerically expensive through Λ; Gram conditioning and tail estimates are physical requirements. Gaussian closure and ergodic homoclinic actions remain extra hypotheses. No generic transform principle supplies them from ordinary mixing.

Focused research directions.

1. **Quantitative saturation in the exact array category.** Specify finitely many complete-data tests, certify their squared-energy deficits, and propagate Proposition 5.1. Any passage to all tests needs a density and truncation budget, not a finite checklist alone.

2. **Stable operator-square extraction.** Combine low-spectrum tail information with Proposition 5.2. The central question is whether the model supplies a usable gap or observable-specific spectral tail; residual smallness alone cannot.

3. **Information bounds for actual source words.** Estimate $I(W;C)$, or an upper bound on it, from an independently specified model. Finite moments only furnish such a bound if a valid entropy-constrained optimization or approximation argument is supplied. The strict distinction between the actual coupling and the product reference law must remain.

4. **Finite-block transport with rates.** Develop measurable exceptional-set and transport-error estimates so that (8.2) is effective. Pointwise cancellation, by itself, does not provide uniform fibre rates.

5. **Certified homoclinic numerics in special models.** Optimize (10.2) subject to a bound on Λ, with interval or exact Gram entries, orthogonal tail controls, and observable-approximation budgets. Compare against convex weights before claiming an advantage.

These are concrete local programs. None is represented as a completed general proof or a promised effective solution of Rokhlin's problem.

14 Verification Record and Conclusion

All eight mathematical source sections, the main file, preamble, operator-square figure source, and bibliography were read. Targeted rereads covered standardness, saturation and rolling, the two operator-square orientations, dense-range polar cancellation, measurable label descent, finite blocks, arbitrary-coupling entropy, and the final positive integral. The separate source audit agrees that no actionable defect was identified, while explicitly withholding certification of the global infinite chain. It is not evidence of independent acceptance by the mathematical community.

The local helper `_audit/check_M280_local.py`, with fixed seed 722280, records its results and source SHA256 fingerprints in `_audit/M280-local-checks.json`. It passed:

- 600 exact rational parity-moment identities and their total-variation checks, including nontrivial full moments with product proper marginals;
- the exact six-cycle Gram identities and signed annihilation;
- 150 exact rational finite-permutation signed-residual inequalities;
- 100 exact nested-projection energy identities;
- 104 exact finite van der Corput inequalities;
- 40 floating-point exponential-moment diagnostics and six binary-channel information diagnostics, including $J(A) = I/n$.

The exponential and logarithmic checks are not rigorous numerical enclosures. Their role is diagnostic; the convexity and relative-entropy proofs in Section 6 are the mathematical justification. Likewise the exact rational checks are finite regressions, not proofs of the source's infinite extension construction. The source fingerprints identify the files read, not an independently verified Git checkout.

The earlier catalogue entry reflected a partial screening pass. This report's full-source reading is broader; it does not retroactively turn screening into validation. Supporting Ram chapters were read directly. External literature reading was bounded as specified in Section 2; no claim is made to have rechecked every reference in the manuscript. Confidential files remained local. Only public titles, identifiers, and bibliographic metadata were used for web research.

Conclusion. M280 is a direct core match to TR-item-313 on the stated applicability scale. The sharp information bound, centered L^1 cancellation extension, Bessel improvement, screening estimates, and physical signed-residual certificates are substantive and explicitly delimited local contributions. The universal mixing claim remains the source's claim, conditional in this report on its complete structural proof. The exact quantifiers, whole-array conditioning, standard-factor reduction, and distinction between actual laws and arbitrary joinings are preserved throughout.

A Source Locators and Reproducibility

The catalogue pins source commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. The source base, relative to `C:/Users/mural/OneDrive/Documents/OpenAI-722-math/`, is

> `_sources/openai-math/preprints/Rokhlins-multiple-mixing-problem-for-one-transformation-September-23-2026/build/`

The filenames below are relative to that base. Labels are the actual source labels, not newly assigned theorem numbers.

File	Principal locators and use
`sections/01-introduction.tex`	`thm:main`; exact claim and quantifiers.
`sections/02-reduction.tex`	`prop:zero-entropy-witness`; `eq:entropy-past-formula`; `eq:witness-correlation`.
`sections/03-arrays.tex`	`def:array-object`, `prop:saturation`, `eq:saturation-equality`, `lem:local-conditioning`, `lem:rolled-extension`, `eq:parent-screening`, `prop:square-locality`, `lem:parameter-geometry`.
`sections/04-operators.tex`	`def:active-space`, `lem:exact-lines`, `lem:rectangle`, `lem:active-reduction`, `lem:fixed-squares`, `lem:polar-group`, `lem:contraction-completeness`, `prop:common-exact-lines`.
`sections/05-labels-blocks.tex`	`lem:active-exact`, `prop:measurable-labels`, `lem:counting-lift`, `lem:finite-blocks`, `prop:block-witness`, `eq:witness-block-transport`.
`sections/06-cancellation.tex`	Dissociated-character entropy proof; `lem:orbit-alternatives`, `lem:polynomial-cancellation`, `prop:orbit-cancellation`.
`sections/07-main-proof.tex`	`sec:main-proof`; positive integral and bounded convergence.
`sections/08-consequences.tex`	`cor:endomorphisms-flows`; natural extension and continuous-time reductions.
`main.tex`, `preamble.tex`, `figures/operator-square.tex`, `references.bib`	Read completely for notation, diagram, assembly, and citations. Reading the bibliography does not mean auditing every cited work.

The exact confidential corpus files are:

1. `C:/Users/mural/OneDrive/Documents/MyDocuments2026/res-rt/res-math/TR-item-313-astra.tex`. Sections *The Exact Problem and Its Quantifiers*; *Two Exact Obstructions to Marginal-Only Closure* (`eq:parity`); *Limits of Actual Off-Diagonal Laws*; *A Quantitative Homoclinic Residual Theorem* (`thm:main`, `eq:recursive`, `cor:all`); *Ram Gram Optimization and Certified Error Transfer* (`eq:soc`, `eq:tailgram`); *A Gaussian Specialization With Covariance Uncertainty* (`eq:gauss`, `eq:gaussblock`). The remaining mathematical body was also read.

2. `C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume21/chapters/ch01d-physical-metrics.tex`. Locators `v21c01:physical-gram`, `v21c01:full-residual`, `v21c01:residual-decomposition`, `v21c01:approximate-gram`, `v21c01:finite-measurements`.

3. `C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume21/chapters/ch02/extremal.tex`. Locators `v21c02:moment-geometry`, `v21c02:dual-equality`, `v21c02:robust-geometry`, `v21c02:robust-dual`.

Local audit files are `_audit/M280-source-audit.txt`, `_audit/check_M280_local.py`, and `_audit/M280-local-checks.json`. The helper is executable with the supplied local Python runtime. It writes only its own audit JSON. No original corpus or manuscript file is modified.

References

[1] OpenAI, *Rokhlin's multiple-mixing problem for one transformation*, September 23, 2026. Pinned manuscript claim, not treated here as established independent truth. https://github.com/openai/math/blob/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/Rokhlins-multiple-mixing-problem-for-one-transformation-September-23-2026/paper.pdf. Source locators are in Appendix A.

[2] M. Subbarao, *Ram Calculus Certificates for Rokhlin's Multiple-Mixing Problem*, Technical Report TR-item-313-astra, Integral Research, confidential local corpus. Exact file and theorem locators: Appendix A.

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[4] M. Subbarao, *Ram Calculus*, Volume 21, `chapters/ch02/extremal.tex`, confidential local corpus. Sections "Finite-Dimensional Geometry of Feasible Data", "Extremal Values of an Unobserved Target", and "Observation Uncertainty and Grid Error".

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